

AS Level Computer Science (9618) — Key Facts & Reference

Cambridge International AS 9618 (Paper 1 & 2). No formulas — this sheet is the recall material: representations, logic, protocols, pseudocode rules.

INFORMATION REPRESENTATION

Binary/hex/denary	hex digit = 4 bits; $2^x = 256$ values in a byte
Two's complement	flip + 1; n-bit range: $2 \cdot \{^1$ to $2 \cdot \{^1$ 1
Binary fractions	bits after the point: $\frac{1}{2}$, $\frac{1}{4}$, ! [... (0.101 = 0.625)
Floating point	sign + mantissa + exponent; more exponent bits = bigger range, less precision
Character sets	ASCII 7-bit; Unicode UTF-8 backward compatible
Error checking	parity (even/odd), checksum, check digit — detect, don't correct (parity block can correct)
Media sizes	image: $w \times h \times \text{depth}$; sound: $\text{rate} \times \text{resolution} \times \text{time} \times \text{channels}$
Compression	lossless (RLE: value,count pairs; dictionary) vs lossy (perceptual)

COMMUNICATION & NETWORKING

Serial vs parallel	serial: one bit at a time, less skew/crosstalk over distance
Direction	simplex (one way), half-duplex (both, alternating), full-duplex (both, simultaneous)
Protocols	HTTP/HTTPS, FTP, SMTP (sending mail), POP3/IMAP (receiving), BitTorrent (P2P)
Circuit vs packet	circuit: dedicated path; packet: routed independently, reassembled
Hardware	router (routes between networks), switch (LAN by MAC), bridge (joins LANs), repeater (boosts signal), NIC, WAP
Internet structure	ISP !' DNS lookup !' IP address; URL = protocol + domain
Client-server vs P2P	server: central services; P2P: each node both client and server

HARDWARE & LOGIC

Von Neumann	stored program: instructions + data in same memory; FDE cycle
Registers	PC (next instruction), MAR (address), MDR (data), ACC (accumulator), CIR (current instruction)
Buses	address (one-way), data + control (two-way)
Interrupts	signal causing current FDE to pause and service a routine
Logic gates	NOT, AND, OR, NAND, NOR, XOR — build the truth table
Boolean algebra	$A \cdot 0 = 0$; $A \cdot 1 = A$; $A + 0 = A$; $A + 1 = 1$; $A \cdot A = A$; $A + A = A$; $(A')' = A$

PROCESSOR FUNDAMENTALS & SYSTEM SOFTWARE

Assembly	one instruction per line: LDM, STO, ADD, INC, DEC, CMP, JMP, JPE, JPN, IN, OUT, END
Addressing modes	immediate #n, direct (address), indirect ((address)), indexed (base+offset)
OS tasks	memory/process/file/device/security management, HCI, interrupts
Translators	assembler (asm!' machine); compiler (all at once, executed by line)
IDE features	editor, debugger (breakpoints, stepping, watch), compiler/interpreter integration

SECURITY, PRIVACY & ETHICS

Threats	hacking, malware (virus, spyware), phishing, pharming, DoS, man-in-the-middle
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Protections	firewall, antivirus, encryption (symmetric AES / asymmetric RSA), access levels, biometrics
Data policies	privacy laws: data must be accurate, secure, collected fairly, rights of access

PSEUDOCODE (Paper 2) — exam-standard constructs

Assign & I/O	<code>x ← 5; OUTPUT x; INPUT x</code>
Selection	<code>IF c THEN ... ELSE ... ENDIF; CASE OF ... ENDCASE</code>
Loops	<code>FOR i ← 1 TO n ... NEXT; WHILE ... ENDWHILE; REPEAT ... UNTIL</code>
Functions	<code>FUNCTION name(params) RETURNS type ... ENDFUNCTION; CALL name()</code>
Arrays & records	<code>DECLARE a[1:100] OF INTEGER; RECORD ... ENDRECORD</code>
File operations	<code>OPENFILE ... FOR READ/WRITE/APPEND; READFILE, WRITEFILE, CLOSEFILE</code>

Paper 2 marks method: declare types, match pseudocode syntax exactly, trace every variable.